



ÓBUDAI EGYETEM
NEUMANN JÁNOS INFORMATIKAI KAR

GEEKDAY 2026

14TH NON-STOP (24 HOURS LONG) PROGRAMMING AI COMPETITION
(MAIN TASK)

TIME SCHEDULE*

*the organizers reserve the right to change anything

2026.04.24. FRIDAY

13:00 – 13:15 OPENING, DESCRIPTION OF TASKS

13:15 – SOLVING PROBLEM(S) (DEVELOPMENT TIME)

2026.04.24. SATURDAY

12:00 PEERS FREEZE / SMALL TASK BONUS FREEZE
(ONE PEER PER TEAM)

13:00 END OF DEVELOPMENT

13:15 – 13:45 CHAMPIONSHIP, ANNOUNCEMENT OF RESULTS

MAIN TASK – FIGHT FIRE WITH RED ALERT

Your task is to create a client application that will **AUTOMATICALLY (WITHOUT ANY USER INTERACTION)** control three kind of units (in the bottom, from left to right: firefighter people, firefighter drones and firefighter trucks), using the OpenRA game engine and gRPC communication. The aim is to put out the fires that are randomly appearing and spreading throughout the map (the small yellow things in the upper-left side of the image).



The primary aim is to extinguish (kill) fires. The secondary is to damage fires. The third tiebreaker number is the distance covered by your units.

The map size and layout can change throughout the competition.

To register, first you need to tell us (in person, quickly) your team name that you want to use in the game (must match the regex `[a-zA-Z0-9_]{3,15}`), which we will manually add to the demo map to enable communication. Also, give us google-compatible email address(es) so that we can share the google drive folder with you (this is where all the announcements will be made).

To communicate with the server, you need to write an **automatic** client to a gRPC server service, the implementation of this automatic client is the main task. One team can develop multiple clients side-by-side, but at the end (in the final competition) only ONE CLIENT/ONE PEER can be active per team.

Grpc unary service: **SayHello** (teamName) ... This command will associate your grpc client peer with the specified team name. There is no IP filter during the development phase, so please be nice and don't steal the units of someone else... At the end (in the final competition) every teamName will be IP bound to your main client IP.

Grpc full-duplex (bi-directional) streaming service: **CommunicateWithStreams** (stream input, + stream output) ... You can find the full exact proto file and the possible Operations in the google drive folder.

Essentially, the server will periodically send to every client the data about their units and also the location of fire/water segments that they see. The client has to respond what the individual units should do.

The exact message formats and game rules will refresh in the google drive folder.

This page WILL refresh continuously, please verify the contents regularly!

<https://drive.google.com/drive/folders/1yAkyX7nXeyMj79hDQ3FWqQhP36A4UDID>

The process of competition:

1. After the opening ceremony the teams receive the tasks and they work 24 hours to solve it (optionally, they also solve the side tasks).
2. During the 24 hours, the side tasks are also evaluated; and the gathered bonuses are added to the server. After that, the championship phase begins which will create the final rankings.
3. In the championship phase, we will do a free-for-all round to determine the final ranking.

Determining the rankings:

- number of extinguished/killed fire units
- amount of damage dealt to fire units
- distance covered by all units

Abilities that can be obtained with the side tasks:

These abilities will be continuously activated: they are not limited in time. Also, the participants can choose to complete any of these tasks in any order (it is acceptable if a team chooses to implement only one side task).

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|--------|--------|---------------------------------------|
| - I. | +Style | - Unique sprites for the player units |
| - II. | +Speed | - Increase the speed of game units |
| - III. | +Sight | - Increase the sight of game units |